# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE CODE:** CSA5101
**COURSE NAME:** Cryptography and Network Security

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**COURSE OUTCOME COVERED**

**CO2:** Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

**Analytical Problem Solving: 40%**

**QUESTIONS:** **Total Marks: 50**

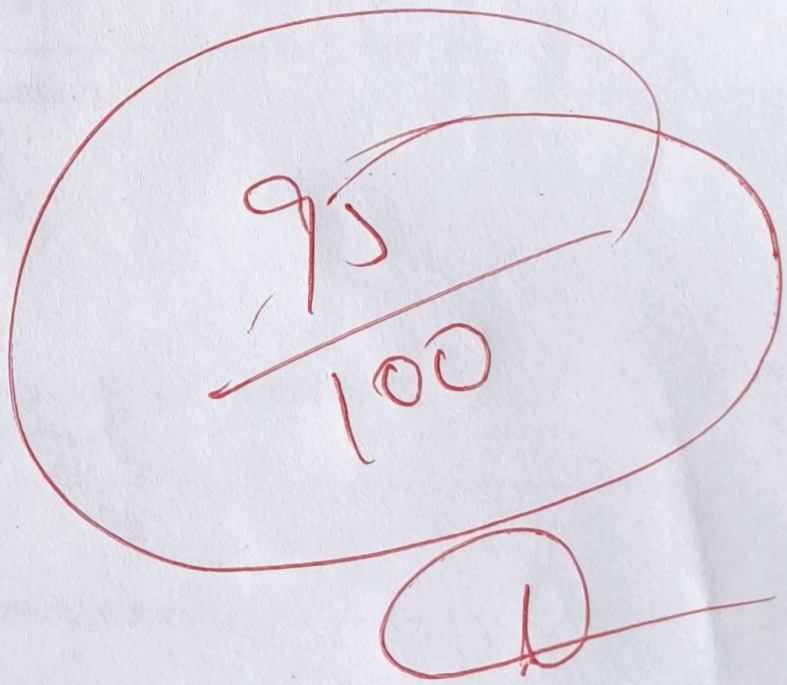
**Annexure A**

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***Code of Conduct:***

*I* S. Madhuri (192465031) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** S. Madhuri



## Annexure A
## Analytical Problem Solving

### 1. RSA Key Generation and Security Analysis
Given:
$p = 17$, $q = 19$
Compute n and $\phi(n)$
Choose $e = 5$ and calculate the private key d
Encrypt the message $M = 7$
Decrypt the ciphertext and verify correctness
Analyze the impact of choosing a small prime number on RSA security

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### 2. ElGamal Encryption Analysis
Given:
$p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$
Compute the public key
Perform encryption to generate ciphertext pair (C1, C2)
Decrypt the ciphertext to retrieve the original message
Analyze how randomness (k) improves security in ElGamal

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### 3. RSA Digital Signature Verification
Given:
$p = 11$, $q = 17$, $e = 7$, message $M = 8$
Compute n and $\phi(n)$
Find private key d
Generate the digital signature for the message
Verify the signature using the public key
Analyze how digital signatures ensure non-repudiation

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### 4. Diffie-Hellman with Attack Resistance Evaluation
Given:
$p = 29$, $g = 2$
User A private key = 5
User B private key = 12
Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

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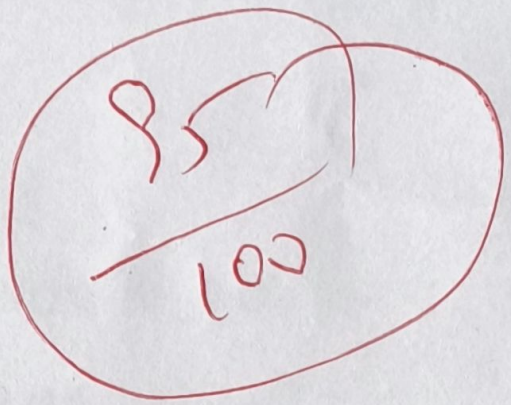
### 5. Hybrid Encryption System Analysis
Given:
A system uses RSA for key exchange and AES for data encryption
RSA encryption time = 5 ms per key exchange
AES encryption time = 0.5 ms per data block
Total data blocks = 500
Calculate total time taken for encryption
Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Problem Understanding | 20% | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection | 20% | Selects most appropriate method/formula with strong justification | Selects correct method with minor errors | Inappropriate or partially correct method | Unable to select method |
| Solution Process | 25% | Logical, systematic steps with correct approach throughout | Mostly correct steps with minor mistakes | Disorganized steps; multiple errors | No clear process |

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|---|---|---|---|---|---|
| Accuracy of Calculation | 20% 20 | All calculations accurate; correct final answer | Minor calculation errors | Frequent errors; incorrect final answer | No valid calculations |
| Interpretation of Results | 15% 15 | Clearly interprets results with units and engineering relevance | Basic interpretation with minor omissions | Limited or unclear interpretation | No interpretation |



1. RSA Key Genration & security analysis:

Given,

First prime number: p=17

Second prime number : q=19

Public exponent : e = 5

Message M=17

Tasks:

1. Compute n & $\phi(n)$.
2. Calculate the private key d.
3. Encrypt the massage M=7.
4. Decrypt the cipher text & verify correctness.
5. Analyze the impact of choosing small prime numbers on RSA security

Introduction

RSA(Rivest-Shamir-Adlman) is an asymmetric cryptographic algorithm that uses a public key for encryption & a private key for decryption. Its security relies on the computational difficulty of factoring a large integer into its prime factors.

Calculation of n & $\phi(n)$.

The RSA modulus is calculated by multiplying the two prime numbers.

$$n = p \times q$$

substituting the values:

$$n = 17 \times 19$$

$$n = 323$$

Eulers totient function is calculated as

$$\phi(n) = (p-1)(q-1)$$

$$= (17-1)(19-1)$$

$$= 16 \times 18$$

$$\phi(n) = 288$$

Therefore, the modulus is 323 & Eulers totient value is 288.

## Calculation of private key d

The private exponent d is the modular multiplicative inverse of the public exponent e.

Formula:

$$ed = 1 \pmod{\phi(n)}$$

Substituting the values:

$$5d = 1 \pmod{288}$$

Using the extended Euclidean algorithm:

$$288 = 5(57) + 3$$

$$5 = 3(1) + 2$$

$$3 = 2(1) + 1$$

Working backword:

$$1 = 3 - 2$$

$$1 = 3 - (5-3)$$

$$1 = 2(3) - 5$$

Substituting $3 = 288 - 5(57)$:

$$1 = 2(288) - 115(5)$$

Thus,

$$-115(5) = 1 \pmod{288}$$

Adding 288 to -115 gives:

$$d = -115 + 288$$

$$d = 173$$

Verification:

$$e \times d = 5 \times 173 = 865$$

$$865 = 3(288) + 1$$

$$\therefore 865 \bmod 288 = 1$$

Hence $d = 173$ is correct.

Public key: $(e, n) = (5, 323)$

Private key: $(d, n) = (173, 323)$

Encryption

$$C = M^e \bmod n$$

$$C = 7^5 \bmod 323$$

$$C = 16807 \bmod 323$$

$$C = 11$$

Decryption:

$$M = c^d \bmod n$$
$$M = 11^{173} \bmod 323$$

Using the Chinese Remainder theorem:

$$M \bmod 17 = 7$$
$$M \bmod 19 = 7$$

$$\therefore M = 7$$

Verification:

Original message = 7

Decrypted message = 7.

Hence, RSA encryption & decryption are successful.

Security analysis:

- Easy factorization: Small primes make n = 323 easy to factor.
- Private-key recovery: An attacker can calculate $\phi(n)$ & recover the private key.
- Reduced confidentiality: Intercepted message can be decrypted.
- Security improvement: Use sufficiently large keys, such as 2048 bit RSA & secure padding such as RSA-OAEP.

Conclusion: RSA successfully encrypts & decrypts the message. However, small prime numbers make it insecure for practical communication.

2 Elgamal Encryption analysis:

Given,

Prime number: p: 23

Genurator: g: 5

Private key : x=7

Random number : k=3

Massage : M=10

Tasks:

1. Compute the public key.
2. Perform encryption to genurate the ciphur text pair $(C_1, C_2)$.
3. Decrypt the ciphertext to retrieve the original message.
4. Analyze how randomness k improves security in E I Gamal.

Introduction:

E1Gamal is an asymmetric cryptographic algorithm based on the mathematical difficulty of the discrete logarithmic problem.

Calculate public key:

$$y = g^x \bmod p$$

$$y = 5^7 \bmod 23$$

$$y = (4 \times 2 \times 5) \bmod 23$$

$y = 17$

public key: (23, 5, 17)

Private key: $x = 7$

Encryption:

ElGamal produces two ciphertext components.

$C_1$:

$$C_1 = g^k \bmod p$$

$$C_1 = 5^3 \bmod 23$$

$$C_1 = 10$$

$C_2$:

$$C_2 = M \times y^k \bmod p$$

$$C_2 = 10 \times 17^3 \bmod 23$$

since,

$$17^3 \bmod 23 = 14$$

$$C_2 = 10 \times 14 \bmod 23$$

$$C_2 = 2$$

$(C_1, C_2) = (10, 2)$.

Decryption:

Calculate the shared secret:

$$S = C_1^x \bmod p$$

$$S = 10^7 \bmod 23$$

$$S = 14$$

Find its modular inverse:

$$14 \times 5 = 70$$

$$70 \bmod 23 = 1$$

$$\therefore 5^{-1} = 5$$

Recover the original message:

$$M = C_2 \times 5^{-1} \bmod p$$

$$= 2 \times 5 \bmod 23$$

$$M = 10$$

Verification:

Original message = 10

Decrypted message = 10.

Hence, ElGamal encryption & decryption are successful.

Security analysis of Randomness k:

- Different ciphertexts: Different random values generally produce different ciphertexts for the same message.
- Unpredictability: A secure random value makes ciphertext prediction more difficult.
- Confidentiality: Randomness helps conceal the original message
- Risk of reuse: Reusing k can reveal relationships b/w messages & weaken security.

• Secure implementation: Generate fresh, unpredictable random values & use sufficiently large cryptographic paramaters.

Conclusion:

ElGamal successfully encrypts the message into (10,2) & recovers the original message 10. Randomness improves confidentiality, but secure paramaters & proper random-number generation are essential for practical security.